

Analysis and Design Report for VStay

Villa Booking System

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Course: SWD392

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May 27, 2026

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1 Introduction

VStay is a villa booking system developed for a software analysis and design assignment. The system allows guests to browse villas, search and filter villa information, view villa details, create bookings, cancel bookings, and make simulated payments. It also provides administration features for managing villas and booking statuses.

The problem comes from small accommodation businesses that often manage villa information, bookings, and payments manually through phone calls, messages, or notes. Without a centralized system, information can become inconsistent and booking records can be difficult to track. VStay focuses on the essential workflows required for an academic software design project.

1.1 Project Summary

Item	Description
Project name	VStay Villa Booking System
Course	SWD392 - Software Design
Main users	Guest and Admin
Main purpose	Analyze, design, and implement a simple villa booking backend prototype for an academic assignment.
Core features	Villa browsing, booking management, payment simulation, villa administration, and booking status management.
Technology stack	Java, Spring Boot, Maven, Spring Data JPA, Spring Security, Swagger/OpenAPI, and H2 demo database.
Repository	https://github.com/kenji-cmyk/mini-booking-villa

1.2 Objectives

- Allow guests to view, search, filter, and inspect villa details.
- Allow guests to create, view, cancel bookings, and make payments.
- Allow admins to create, update, delete, and view villa records.
- Allow admins to view all bookings and update booking statuses.
- Provide complete analysis and design artifacts, including requirements, UML diagrams, design patterns, and an implementation prototype.

1.3 Scope

The project scope is intentionally limited to core villa booking features, villa management, booking management, and simplified payment handling. Advanced commercial features such as AI recommendations, loyalty programs, chat, multi-vendor marketplace support, mobile applications, dynamic pricing, and advanced financial reports are out of scope.

2 System Requirements

2.1 Actors

Actor	Description
Guest	A user who browses villas, searches and filters villas, creates bookings, cancels bookings, makes payments, and checks booking or payment status.
Admin	A system user responsible for managing villa information and updating booking status.

2.2 Functional Requirements

ID	Requirement Summary
FR-01–FR-04	Guests can view the villa list, search, filter, and view villa details.
FR-05–FR-09	Guests can create bookings, while the system validates data, checks availability, shows booking details, and allows eligible cancellation.
FR-10–FR-12	Guests can submit a simulated payment, the system records payment information, and guests can view payment status.
FR-13–FR-16	Admins can create, update, delete, and view villa records. Villas with active bookings cannot be deleted.
FR-17–FR-20	Admins can view all bookings, view booking details, and update booking status according to valid business rules.

2.3 Non-Functional Requirements

- **Performance:** villa lists, search results, and booking details should respond within about 3 seconds for demo data.
- **Security:** admin functions must be restricted to authorized admin users; sensitive payment data must not be fully exposed.
- **Reliability:** the system must prevent duplicate active bookings for the same villa and overlapping dates.
- **Maintainability:** the backend must clearly separate controller, service, repository, entity, DTO, mapper, payment, and exception responsibilities.
- **Validation:** required fields, booking dates, guest count, and payment information must be validated at system boundaries.

3 Use Case Diagram

3.1 AI Prompt Used

Create a UML use case diagram for a villa booking management system.

Requirements:

- Guests can view villas, search villas, view villa details, create bookings, cancel bookings, make payments, and view payment status.
- Admins can create, update, delete, and manage villas and bookings.

Generate a PlantUML use case diagram using Guest and Admin actors.
Keep the diagram simple and suitable for a university software engineering assignment.

3.2 UML Code

```
@startuml
left to right direction

actor Guest
actor Admin

rectangle "VStay Villa Booking System" {

    usecase "UC-01\nView Villas" as UC01
    usecase "UC-02\nSearch and Filter Villas" as UC02
    usecase "UC-03\nView Villa Details" as UC03

    usecase "UC-04\nCreate Booking" as UC04
    usecase "Validate Booking Information" as UC04A
    usecase "Check Villa Availability" as UC04B

    usecase "UC-05\nView Booking Details" as UC05
    usecase "UC-06\nCancel Booking" as UC06
    usecase "UC-07\nMake Payment" as UC07
    usecase "UC-08\nView Payment Status" as UC08

    usecase "UC-09\nCreate Villa" as UC09
    usecase "UC-10\nUpdate Villa" as UC10
    usecase "UC-11\nDelete Villa" as UC11
    usecase "UC-12\nView Villa List" as UC12
    usecase "UC-13\nView All Bookings" as UC13
    usecase "UC-14\nUpdate Booking Status" as UC14
}

Guest --> UC01
Guest --> UC02
Guest --> UC03
Guest --> UC04
Guest --> UC05
Guest --> UC06
Guest --> UC07
Guest --> UC08

Admin --> UC09
Admin --> UC10
Admin --> UC11
Admin --> UC12
Admin --> UC13
Admin --> UC14
```

```
UC04 ..> UC04A : <<include>>
UC04 ..> UC04B : <<include>>
```

```
@enduml
```

3.3 UML Image

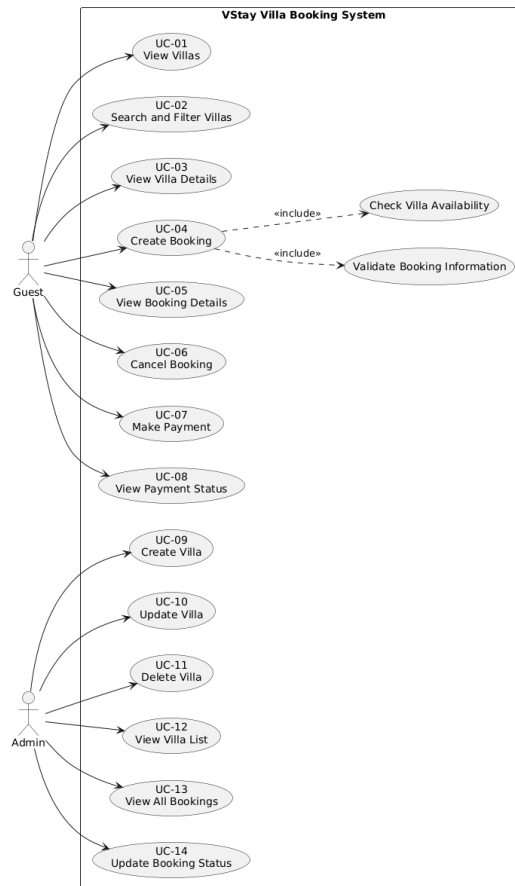


Figure 1: Use Case Diagram of VStay

3.4 Short Explanation

The use case diagram shows two main actors: Guest and Admin. Guests handle villa browsing, booking, cancellation, and payment actions. Admins manage villa records and booking statuses. The Create Booking use case includes validation and availability checking because these are mandatory business rules.

4 Package Diagram

4.1 AI Prompt Used

```
Create a UML package diagram for a Spring Boot villa booking management system.
```

The system includes villa browsing, booking management, payment processing using MomoPay and VNPay, and admin management.
Use a layered architecture with controller, service, payment, repository, entity, enums, dto, mapper, exception, and config packages.
Include Strategy Pattern and Factory Pattern for payment processing.

4.2 UML Code

```
@startuml
skinparam packageStyle rectangle
skinparam shadowing false
skinparam linetype ortho

package "vstay" {

    package "controller" {

        class VillaController
        class BookingController
        class PaymentController

        class AdminVillaController
        class AdminBookingController
    }

    package "service" {

        class VillaService
        class BookingService
        class PaymentService
    }

    package "payment" {

        class PaymentStrategyFactory

        interface PaymentStrategy

        class MomoPaymentStrategy
        class VNPayPaymentStrategy
    }

    package "repository" {

        interface VillaRepository
        interface BookingRepository
        interface PaymentRepository
        interface UserRepository
    }

    package "entity" {
```

```

class User
class Villa
class Booking
class Payment
}

package "enums" {

    enum Role
    enum BookingStatus
    enum PaymentStatus
}

package "dto" {

    class VillaResponse
    class BookingRequest
    class BookingResponse

    class PaymentRequest
    class PaymentResponse

    class CreateVillaRequest
    class UpdateVillaRequest
}

package "mapper" {

    class VillaMapper
    class BookingMapper
    class PaymentMapper
}

package "exception" {

    class NotFoundException
    class ValidationException
    class BookingConflictException
    class AccessDeniedException
}

package "config" {

    class SecurityConfig
}
}

, =====
, CONTROLLER -> SERVICE
, =====

```



```

VillaController --> VillaService
BookingController --> BookingService
PaymentController --> PaymentService

AdminVillaController --> VillaService
AdminBookingController --> BookingService

' =====
' SERVICE -> REPOSITORY
' =====

VillaService --> VillaRepository
BookingService --> BookingRepository
PaymentService --> PaymentRepository
BookingService --> VillaRepository

' =====
' SERVICE -> ENTITY
' =====

VillaService --> Villa
BookingService --> Booking
PaymentService --> Payment

' =====
' DTO RELATIONSHIPS
' =====

VillaController --> VillaResponse

BookingController --> BookingRequest
BookingController --> BookingResponse

PaymentController --> PaymentRequest
PaymentController --> PaymentResponse

AdminVillaController --> CreateVillaRequest
AdminVillaController --> UpdateVillaRequest

' =====
' MAPPER RELATIONSHIPS
' =====

VillaService --> VillaMapper
BookingService --> BookingMapper
PaymentService --> PaymentMapper

VillaMapper --> Villa
BookingMapper --> Booking
PaymentMapper --> Payment

```

```

' =====
' STRATEGY PATTERN
' =====

PaymentService --> PaymentStrategyFactory

PaymentStrategyFactory --> PaymentStrategy

PaymentStrategy <|.. MomoPaymentStrategy
PaymentStrategy <|.. VNPayPaymentStrategy

PaymentService --> PaymentStrategy

' =====
' EXCEPTION HANDLING
' =====

VillaService --> ValidationException
BookingService --> BookingConflictException
PaymentService --> ValidationException

' =====
' SECURITY
' =====

SecurityConfig --> Role
User --> Role

Booking --> BookingStatus
Payment --> PaymentStatus

@enduml

```

4.3 UML Image

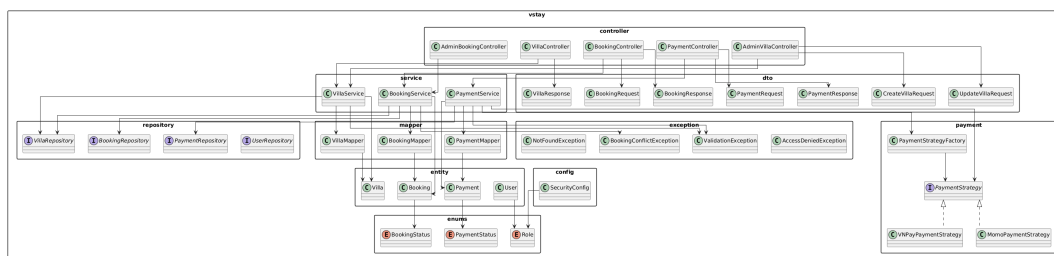


Figure 2: Package Diagram of the VStay Backend

4.4 Short Explanation

The package diagram describes the Spring Boot backend using a layered structure. Controllers receive API requests, services handle business logic, repositories access data, entities and enums represent the domain model, DTOs and mappers control API data

transfer, exceptions centralize error handling, and config stores security configuration. The payment package applies Strategy and Factory patterns for provider-based payment processing.

5 Class Diagram

5.1 AI Prompt Used

Create a simple UML class diagram for a villa booking management system.

The system includes guests and admins, villas, bookings, payments, booking status, and payment status.

Only include the main domain classes, their basic attributes, simple methods, enum values, and relationships.

Do not include technical implementation layers.

5.2 UML Code

```
@startuml
skinparam classAttributeIconSize 0
skinparam shadowing false

class User {
    - id: Long
    - fullName: String
    - email: String
    - phone: String
    - role: Role
    + isAdmin(): boolean
    + isGuest(): boolean
}

class Villa {
    - id: Long
    - name: String
    - location: String
    - description: String
    - pricePerNight: BigDecimal
    - capacity: int
    - active: boolean
    + updateInfo(): void
    + deactivate(): void
}

class Booking {
    - id: Long
    - checkInDate: LocalDate
    - checkOutDate: LocalDate
    - numberOfGuests: int
}
```

```

- contactName: String
- contactPhone: String
- status: BookingStatus
+ confirm(): void
+ cancel(): void
+ complete(): void
+ isActive(): boolean
}

class Payment {
- id: Long
- provider: String
- amount: BigDecimal
- status: PaymentStatus
- transactionCode: String
- paidAt: LocalDateTime
+ markSuccessful(): void
+ markFailed(): void
}

enum Role {
    GUEST
    ADMIN
}

enum BookingStatus {
    PENDING
    CONFIRMED
    CANCELLED
    COMPLETED
}

enum PaymentStatus {
    PENDING
    SUCCESSFUL
    FAILED
}

User "1" --> "0..*" Booking : creates
Villa "1" --> "0..*" Booking : is booked in
Booking "1" --> "0..1" Payment : has

User --> Role
Booking --> BookingStatus
Payment --> PaymentStatus

@enduml

```

5.3 UML Image

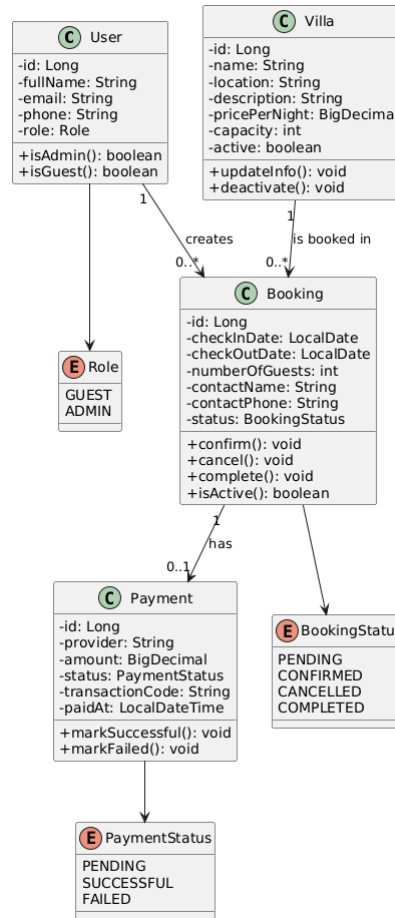


Figure 3: Core Domain Class Diagram

5.4 Short Explanation

The class diagram focuses on four core domain classes: User, Villa, Booking, and Payment. User represents both guests and admins through the Role enum. Villa stores villa information and active status. Booking connects a user to a villa for a date range and tracks booking status. Payment belongs to one booking and stores the simplified payment result.

6 Sequence Diagrams

The sequence diagrams are separated by workflow to keep each behavior readable and aligned with the use cases.

6.1 AI Prompt Used

Create simple UML sequence diagrams for a villa booking management system.

Actors:
- Guest

- Admin

Separate the sequence diagrams by action:

- Guest browses, searches, and views villas
- Guest creates a booking
- Guest cancels a booking
- Guest makes a payment and views payment status
- Admin manages villas
- Admin views bookings and updates booking status

Keep each diagram high-level and business-focused.

Do not include technical implementation layers.

6.2 Browse and View Villas

```
@startuml
skinparam shadowing false
skinparam sequenceMessageAlign center
skinparam responseMessageBelowArrow true

actor Guest
participant "VStay System" as System
database "Stored Data" as Data

Guest -> System: Request villa list or search result
System -> Data: Load matching villas
Data --> System: Villa list
System --> Guest: Show villas

Guest -> System: Select villa
System -> Data: Load villa details
Data --> System: Villa details
System --> Guest: Show villa details

@enduml
```

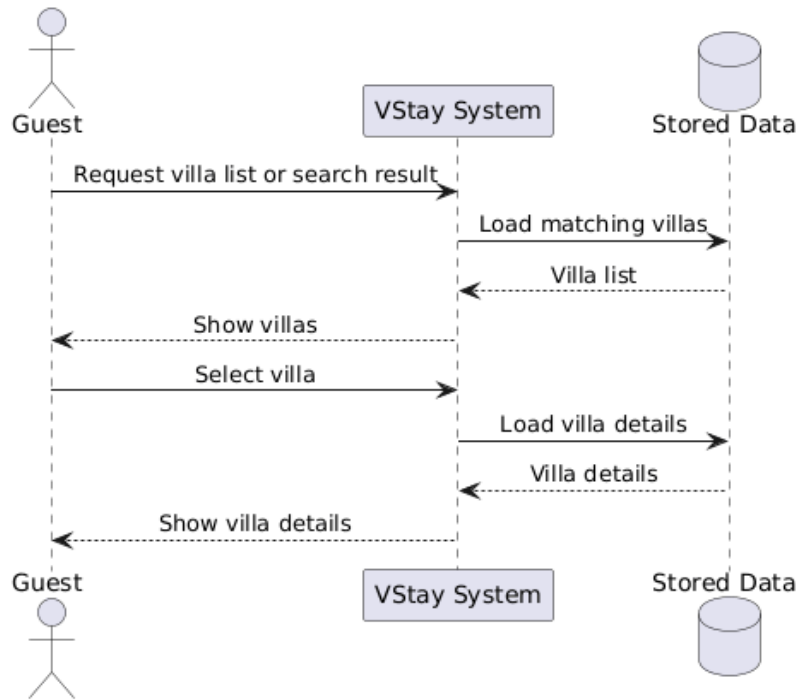


Figure 4: Sequence Diagram for Browsing and Viewing Villas

The guest requests a villa list or search result. The system loads matching villas and returns the list or selected villa details.

6.3 Create Booking

```

@startuml
skinparam shadowing false
skinparam sequenceMessageAlign center
skinparam responseMessageBelowArrow true

actor Guest
participant "VStay System" as System
database "Stored Data" as Data

Guest -> System: Submit booking information
System -> System: Validate dates, guest count, and contact information
System -> Data: Check villa availability
Data --> System: Availability result

alt Villa is not available
    System --> Guest: Show booking conflict message
else Villa is available
    System -> Data: Save booking with PENDING status
    Data --> System: Saved booking
    System --> Guest: Show booking details
end

@enduml
  
```

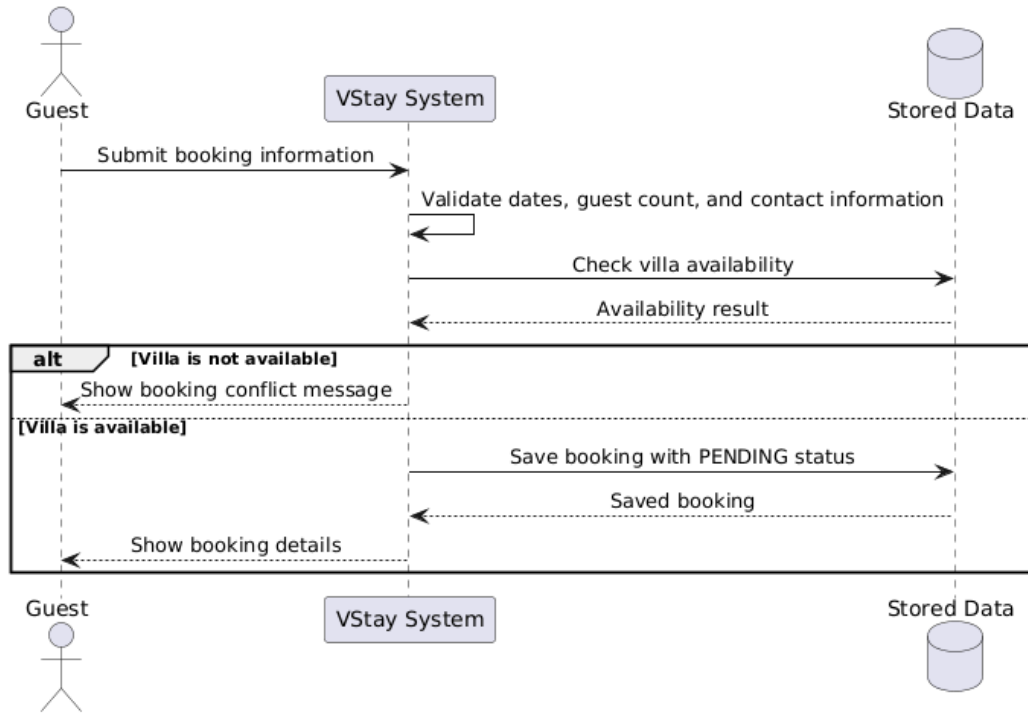


Figure 5: Sequence Diagram for Creating a Booking

The guest submits booking information. The system validates dates, guest count, contact information, and villa availability. If valid, the booking is saved with PENDING status.

6.4 Cancel Booking

```

@startuml
skinparam shadowing false
skinparam sequenceMessageAlign center
skinparam responseMessageBelowArrow true

actor Guest
participant "VStay System" as System
database "Stored Data" as Data

Guest -> System: Request booking cancellation
System -> Data: Load booking
Data --> System: Booking details
System -> System: Check cancellation eligibility

alt Booking can be cancelled
    System -> Data: Update booking status to CANCELLED
    Data --> System: Updated booking
    System --> Guest: Show cancelled booking
else Booking cannot be cancelled
    System --> Guest: Show validation message
end

@enduml
  
```

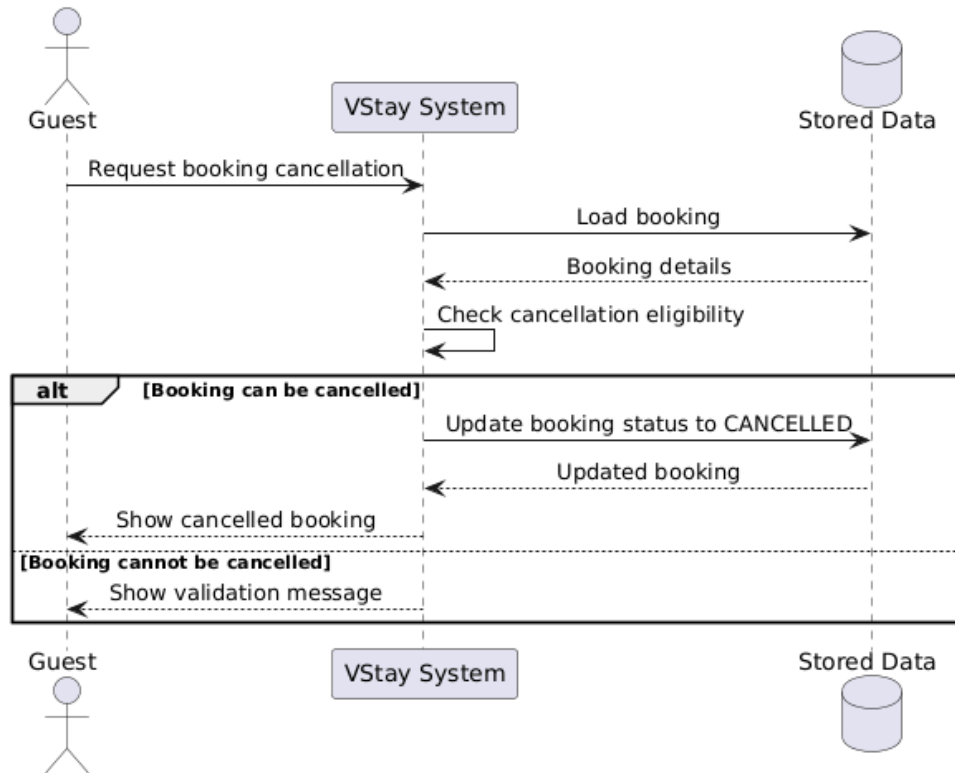



Figure 6: Sequence Diagram for Cancelling a Booking

The guest requests booking cancellation. The system loads the booking, checks ownership and cancellation eligibility, then updates the status to CANCELLED if the request is valid.

6.5 Make Payment and View Payment Status

```

@startuml
skinparam shadowing false
skinparam sequenceMessageAlign center
skinparam responseMessageBelowArrow true

actor Guest
participant "VStay System" as System
database "Stored Data" as Data

Guest -> System: Submit payment information
System -> Data: Load booking
Data --> System: Booking details
System -> System: Process simplified payment

alt Payment is successful
    System -> Data: Save payment with SUCCESSFUL status
    System -> Data: Update booking status to CONFIRMED
    Data --> System: Updated payment and booking
    System --> Guest: Show successful payment status
else Payment fails or is pending
    System -> Data: Save payment with FAILED or PENDING status
    Data --> System: Saved payment
end

```

```

System --> Guest: Show payment status
end

```

```

Guest -> System: Request payment status
System -> Data: Load payment by booking
Data --> System: Payment status
System --> Guest: Show payment status

```

```

@enduml

```

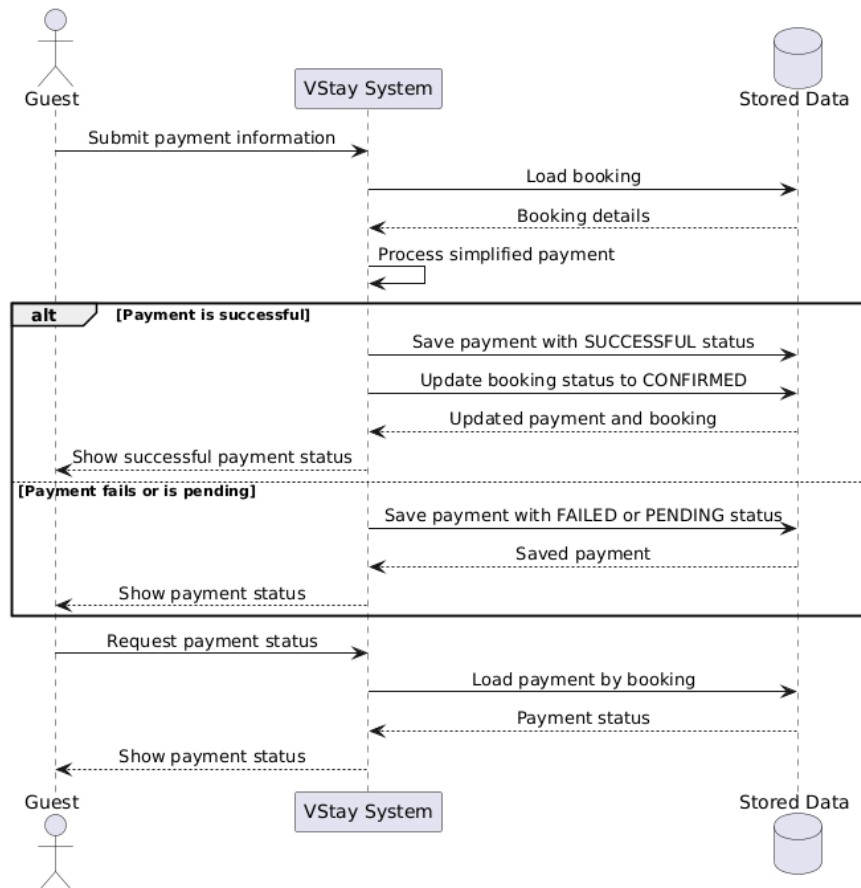


Figure 7: Sequence Diagram for Payment and Payment Status

The guest submits payment for a booking. The system selects a payment strategy, records the payment result, and confirms the booking if the payment succeeds. The guest can also request the payment status later.

6.6 Admin Manage Villas

```

@startuml
skinparam shadowing false
skinparam sequenceMessageAlign center
skinparam responseMessageBelowArrow true

actor Admin
participant "VStay System" as System

```

```

database "Stored Data" as Data

Admin -> System: Create or update villa information
System -> System: Validate villa information
System -> Data: Save villa information
Data --> System: Saved villa
System --> Admin: Show updated villa list

Admin -> System: Delete villa
System -> Data: Check active bookings for villa
Data --> System: Active booking result

alt Villa has active booking
    System --> Admin: Reject villa deletion
else Villa has no active booking
    System -> Data: Delete villa
    Data --> System: Villa deleted
    System --> Admin: Show updated villa list
end

@enduml

```

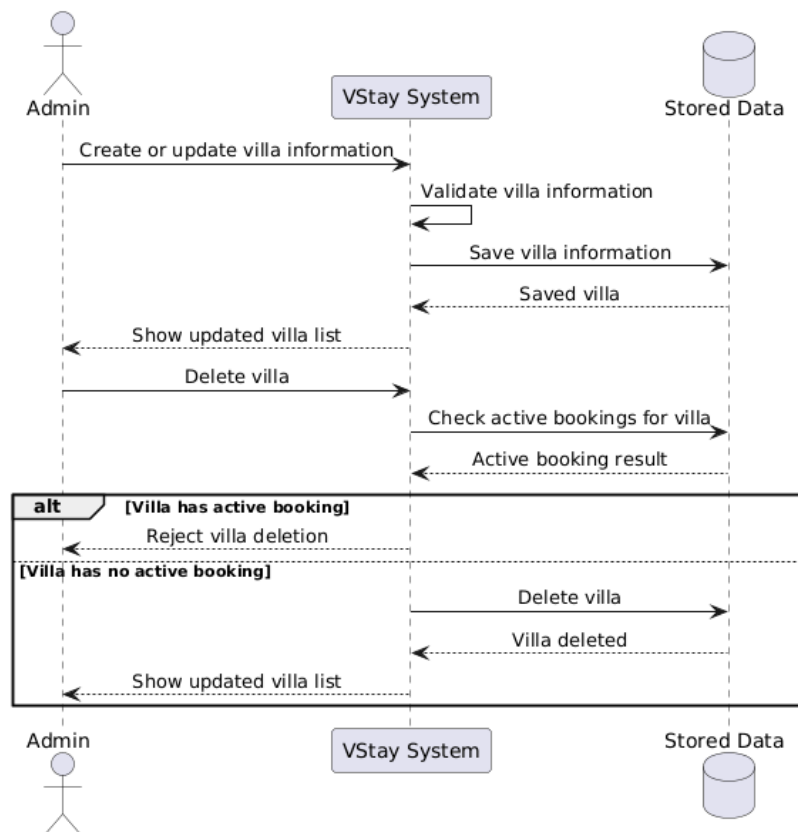


Figure 8: Sequence Diagram for Admin Villa Management

The admin creates or updates villa information after validation. When deleting a villa, the system checks for active bookings and rejects deletion if active bookings still exist.

6.7 Admin Manage Bookings

```
@startuml
skinparam shadowing false
skinparam sequenceMessageAlign center
skinparam responseMessageBelowArrow true

actor Admin
participant "VStay System" as System
database "Stored Data" as Data

Admin -> System: Request all bookings
System -> Data: Load bookings
Data --> System: Booking list
System --> Admin: Show booking list

Admin -> System: Update booking status
System -> System: Validate status change
System -> Data: Save updated booking status
Data --> System: Updated booking
System --> Admin: Show updated booking

@enduml
```

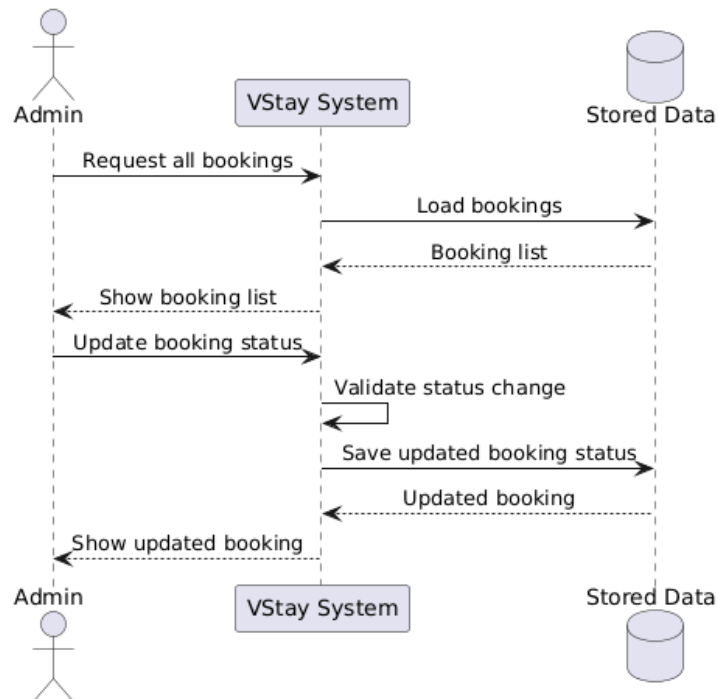


Figure 9: Sequence Diagram for Admin Booking Management

The admin views all bookings and updates booking status. The system validates status transitions to keep booking state consistent.

7 Design Patterns

Pattern	Applied In	Purpose
Layered Architecture	Controller, service, repository, and entity packages	Separates API handling, business logic, data access, and domain representation.
Repository Pattern	Repository interfaces	Hides database query details and allows services to work through clean repository interfaces.
DTO & Mapper Pattern	DTO and mapper packages	Prevents direct exposure of entities through APIs and centralizes request/response conversion.
Strategy Pattern	Payment strategy classes	Separates payment behavior by provider and makes it easier to add new providers.
Factory Pattern	Payment strategy factory	Selects the correct payment strategy by provider and keeps the payment service decoupled from concrete classes.

7.1 Design Pattern UML Note

PlantUML files for the design patterns are available in `docs/uml`. The relevant files are:

- `design-pattern-layered-architecture.puml`
- `design-pattern-repository.puml`
- `design-pattern-dto-mapper.puml`
- `design-pattern-strategy.puml`
- `design-pattern-factory.puml`

8 Code Implementation

The backend prototype is implemented with Java Spring Boot, Maven, Spring Data JPA, Spring Security, Swagger/OpenAPI, and an H2 demo database. The main source code is located in `backend/src/main/java/com/kna/vstay`.

- GitHub repository: <https://github.com/kenji-cmyk/mini-booking-villa>
- Public APIs: `/api/villas`, `/api/bookings`, `/api/payments`
- Admin APIs: `/api/admin/villas`, `/api/admin/bookings`
- Local Swagger UI: `http://localhost:8080/swagger-ui.html`

8.1 Implementation Status

The implementation covers the main use cases from UC-01 to UC-14. Important rules are implemented, including overlapping booking prevention, guest count validation, guest ownership checks for booking and payment actions, payment statuses PENDING, SUCCESSFUL, and FAILED, and valid booking status transitions.

9 Conclusion

VStay satisfies the assignment scope with requirements, use cases, UML diagrams, design patterns, and a backend implementation prototype. The system uses layered architecture, repository, DTO/mapper, strategy, and factory patterns to keep the design clear, maintainable, and suitable for academic demonstration. The main Guest and Admin workflows are described through sequence diagrams and implemented in the backend prototype.

10 References

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